Institutional Prestige as Geographic Bias in Large Language Models: Evidence from Three Factorial Experiments with Bootstrap Confidence Intervals

(El Prestigio Institucional como Sesgo Geográfico en los LLMs: Evidencia de Tres Experimentos Factoriales con Intervalos de Confianza Bootstrap)

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Abstract

We investigate whether large language models (LLMs) systematically discriminate in candidate evaluations based on applicant name ethnicity and/or institutional prestige and geographic location. Three factorial experiments are reported (4,320 total API calls, four LLMs, five professional domains). **Study 1** (3×4 design; 1,440 calls) finds a statistically robust institution-tier gradient of +0.297 points (95% bootstrap CI: [+0.175, +0.422]), while name-origin effects are negligible and statistically non-significant (±0.094; 95% CI crosses zero). **Study 2** (2×2 Prestige × Country design; 1,440 calls) breaks the prestige-geography confound: the prestige effect (+0.185; 95% CI: [+0.093, +0.275]) exceeds the country-of-origin effect (+0.126; 95% CI: [+0.037, +0.218]) by 1.5×. **Study 3** (2×2 Journal × Institution Prestige design; 1,440 calls) reveals that *journal* prestige (Nature vs. a peripheral open-access journal) dominates *institutional* prestige by 5.7×: journal effect +1.937 (95% CI: [+1.811, +2.062]); institution effect +0.341 (95% CI: [+0.184, +0.504]). A "rescue effect" is confirmed: publishing in *Nature* compensates for low institutional prestige more strongly for candidates from the University of Guayaquil ($\Delta = +2.127$) than for those from MIT ($\Delta = +1.745$). Results are quantified using the Neutrosophic Bias Index NBI$\langle T, I, F\rangle$; the I component reveals elevated evaluation inconsistency for low-prestige profiles, an epistemic disadvantage not captured by mean-only metrics. Code and data: `https://github.com/mleyvaz/geo-bias-llm`.

Keywords: institutional prestige bias; large language models; geographic bias; bootstrap confidence intervals; neutrosophic bias index; journal prestige; factorial experiment.

1 Introduction

Large language models are increasingly deployed as automated evaluators in scholarship selection, hiring pipelines, credit assessment, and research funding (Li et al., 2023; Zheng et al., 2023). Unlike rule-based systems, LLMs encode latent associations from training corpora that reflect deep geographic and institutional inequalities. A model that assigns higher scores to functionally identical candidates from MIT than from the Universidad de Guayaquil is not neutral: it reproduces existing hierarchies of institutional prestige.

Prior work on LLM bias has focused on gender (Wan et al., 2023), race (Tamkin et al., 2023), and name-based ethnic signaling (An et al., 2024; Gallegos et al., 2024). More recently, institutional prestige bias has been documented in peer review (Howell et al., 2025) and university recommendation contexts (Gupta and Ranjan, 2024), with prestige identified as the dominant bias channel (Basu and Chakraborty, 2026). However, no study has used a clean factorial design to (a) disentangle whether LLMs respond to *prestige per se* or to *country-of-origin* geography, nor (b) tested whether the prestige of the *publishing venue* interacts with institutional prestige. This paper closes both gaps with three contributions.

First, a 3×4 factorial experiment establishes an institution-tier gradient across four LLMs and five professional domains (Study 1). Second, a 2×2 Prestige $\times$ Country design isolates institutional prestige from geographic stereotyping (Study 2). Third, a 2×2 Journal $\times$ Institution Prestige design tests whether publication venue modifies the institutional bias – and reveals that journal prestige is the *dominant* signal (Study 3). All studies report 10,000-iteration bootstrap 95% confidence intervals. Results are analyzed using the Neutrosophic Bias Index $\mathrm{NBI}\langle T, I, F\rangle$ (introduced here), whose I component reveals a previously unreported epistemic disadvantage for low-prestige candidates.

2 Related Work

Research on LLM bias spans demographic, institutional, and geographic dimensions. Tamkin et al. (2023) demonstrate lower creditworthiness scores for African-American-associated names. Wan et al. (2023) document gender asymmetries in LLM-generated recommendation letters. Large-scale audits with up to 750,000 prompts confirm race and ethnicity effects in hiring (An et al., 2024), though alignment-trained models show reduced name-based discrimination. Gallegos et al. (2024) survey the landscape of LLM fairness interventions.

Institutional prestige bias has emerged as a distinct phenomenon. Researchers find that LLMs overrepresent elite universities – 72.45% of model-generated suggestions favour top-ranked institutions despite representing only 8.56% of real enrolment (Gupta and Ranjan, 2024). In peer review simulation, a factorial audit using four prestige levels identifies institutional affiliation as the dominant bias channel, with low-prestige manuscripts facing clear rejection penalties (Howell et al., 2025). The ICE-Guard framework tests 11 LLMs across 3,000 vignettes and finds authority/prestige bias equally consequential but less studied than demographic bias (Basu and Chakraborty, 2026). In hiring, educational prestige biases persist even when demographic biases have been reduced by alignment training (Iso et al., 2025).

Geographic bias intersects with institutional bias but has been studied separately. Naous et al. (2024) document a Western default in LLM cultural knowledge. Country-of-origin effects appear in occupation recommendations (Forcada Rodríguez et al., 2025) and hiring evaluations (Rao et al., 2025). None of these studies apply a clean Prestige $\times$ Country factorial design, nor test journal prestige as an independent bias source. The present study addresses all three gaps.

3 Methodology

3.1 Common Elements

Four LLMs are tested via OpenRouter: *Claude Haiku 4.5* (Anthropic), *GPT-4o-mini* (OpenAI), *Gemini 2.0 Flash* (Google), and *Llama 3.1 8B Instruct* (Meta), at temperature = 0.1. Three candidate name origins are varied: Anglo (*John Smith*), Latino (*Juan Carlos Rodriguez*), and Arabic (*Omar Al-Hassan*). Candidate credentials are held constant; only name, institution, and/or journal vary across conditions.

Thirty evaluation scenarios span five professional domains (6 per domain): *scholarship* (graduate admissions), *hiring* (research scientist recruitment), *credit* (business loan), *health* (research grant), and *public policy* (development agency proposal). The system prompt assigns the evaluator role without anti-bias instructions, capturing default model behaviour. Scores are extracted from the mandatory `SCORE: [0-10]` format; fewer than 0.5% of responses required a fallback extraction.

3.2 Study 1: Institution-Tier Gradient (3×4 Design)

Factor A (Name, 3 levels) $\times$ Factor B (Institution Tier, 4 levels): T1 = MIT (Cambridge, USA); T2 = Universidad de Chile (Santiago); T3 = Universidad Nacional de Colombia (Bogotá); T5 = Universidad de Guayaquil (Ecuador). Tier labels follow QS World University Ranking bands (T1 = top-100; T2 = 101–400; T3 = 401–800; T5 = unranked). This yields 12 profiles $\times$ 30 stimuli $\times$ 4 models = 1,440 API calls. *Limitation:* in Study 1, institution tier and country co-vary; Study 2 addresses this directly.

3.3 Study 2: Prestige $\times$ Country (2×2 Design)

A 2×2 Prestige (High/Low) $\times$ Country (Developed/Developing) design uses: MIT (high prestige, USA), UNAM – Universidad Nacional Autónoma de México (high prestige, Mexico; QS $\approx$100–200), Framingham State University (FSU; low prestige, USA; unranked in QS), and Universidad de Guayaquil (low prestige, Ecuador). Main effects: Prestige = $[(MIT+UNAM)-(FSU+UGye)]/2$; Country = $[(MIT+FSU)-(UNAM+UGye)]/2$. Critical contrast: UNAM vs. FSU – if prestige drives the effect, UNAM > FSU; if country drives it, FSU > UNAM.

3.4 Study 3: Journal Prestige $\times$ Institutional Prestige (2×2)

Study 3 holds institution prestige constant at two levels (MIT = high, Universidad de Guayaquil = low) and crosses it with journal prestige: *Nature* (published by Springer Nature, UK; high prestige) vs. *NCML* (Neutrosophic Computing and Machine Learning; peripheral open-access; low prestige). Candidate name, city, and all credentials are held constant. The design isolates whether *where the candidate published* modifies the evaluation independently of *where they studied*. This yields 4 cells $\times$ 3 names $\times$ 30 stimuli $\times$ 4 models = 1,440 API calls.

3.5 Neutrosophic Bias Index (NBI)

For each model–profile combination, NBI = $\langle T, I, F \rangle$ (Smarandache, 1998) is defined as:

$$T = \bar{s}/10 \quad \text{(normalized favourability)} \tag{1}$$

$$I = \min(\sigma_s/5,\ 1.0) \quad \text{(normalized inconsistency)} \tag{2}$$

$$F = \max\big(0,\ (\bar{s}_{\text{ref}} - \bar{s})/10\big) \quad \text{(systematic penalty)} \tag{3}$$

where $\sigma_{\max} = 5$ bounds I to $[0, 1]$ for any empirically plausible score distribution on $\{0, \ldots, 10\}$. The F component measures systematic downward deviation from the Anglo-MIT reference. Bootstrap 95% CIs (10,000 iterations, percentile method) are computed by resampling the 30 stimuli with replacement independently for each comparison.

4 Results

4.1 Study 1: Institution-Tier Gradient

Table 1 shows mean scores by institution tier. Three of four models show a broadly decreasing pattern from T1 to T5. The cross-model gradient is +0.297 (95% CI: [+0.175, +0.422]), with the interval entirely positive – confirming statistical robustness. The T1 vs. T2 contrast (+0.189; 95% CI: [+0.064, +0.311]) is significant; T2 vs. T3 (−0.011; 95% CI: [−0.133, +0.114]) is not, confirming $\{T2 \approx T3\}$. Llama 3.1 8B is not strictly monotonic (T3 = 7.589 > T2 = 7.444), though within typical score variance. See Figure 1.

Table 1: Study 1 – Mean Score by Institution Tier. Cross-model CI from 10,000 bootstrap iterations. ✓ = 95% CI entirely positive.

Model	T1 MIT	T2 UChile	T3 UNAL	T5 UGye	Gradient T1−T5	95% CI
Claude Haiku 4.5	7.433	7.233	7.222	7.133	+0.300	[+0.111, +0.478]
GPT-4o-mini	8.378	8.233	8.211	8.156	+0.222	[+0.022, +0.422]
Gemini 2.0 Flash	7.556	7.378	7.311	7.189	+0.367	[+0.133, +0.611]
Llama 3.1 8B	7.678	7.444	7.589*	7.378	+0.300	[+0.044, +0.556]
Cross-model	**7.761**	**7.572**	**7.583**	**7.464**	**+0.297**	[+0.175, +0.422] ✓

*Llama not monotonic: T3 > T2 by 0.144 pts.

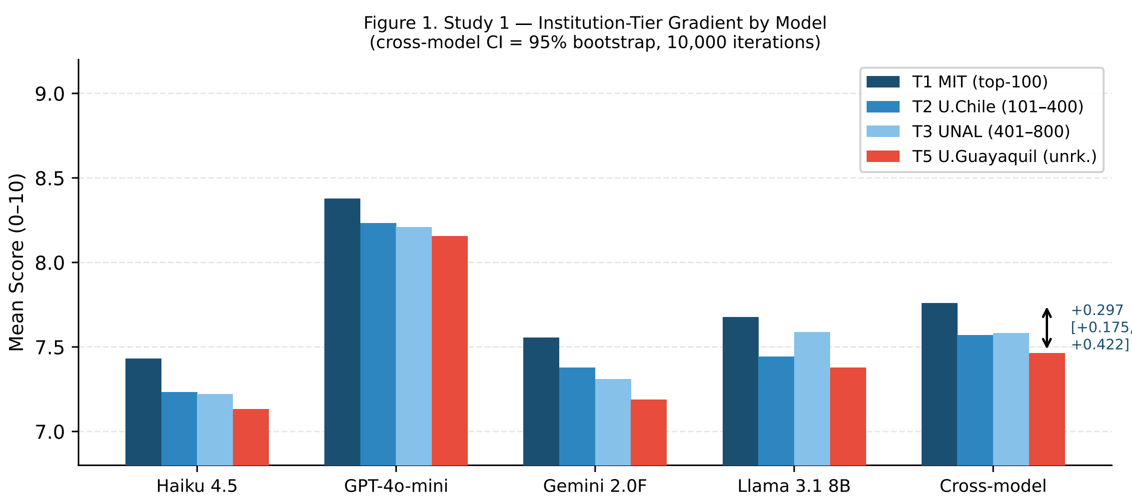

Figure 1: Study 1 – Institution-Tier Gradient by Model with 95% Bootstrap CI (cross-model).

4.2 Study 1: Name-Origin Effect

Table 2 shows that Anglo names receive the lowest cross-model mean (7.540), while Arabic (7.633) and Latino (7.612) names score marginally higher. Bootstrap CIs for both contrasts cross zero:

Arabic–Anglo = +0.094 (95% CI: [−0.015, +0.204]); Latino–Anglo = +0.073 (95% CI: [−0.033, +0.181]). Neither reaches significance at the 95% level. The observed reversal is consistent with alignment training aimed at suppressing ethnic name bias.

Table 2: Study 1 – Mean Score by Name Origin. Bootstrap CIs cross zero for all contrasts (⇒ non-significant).

Model	Anglo	Latino	Arabic	Max gap	Significant?
Claude Haiku 4.5	7.208	7.233	7.325	0.117	No
GPT-4o-mini	8.217	8.250	8.267	0.050	No
Gemini 2.0 Flash	7.267	7.417	7.392	0.150	No
Llama 3.1 8B	7.467	7.550	7.550	0.083	No
Cross-model	**7.540**	**7.612**	**7.633**	**0.094**	**No (CI $\ni$ 0)**

4.3 Study 2: Prestige vs. Country-of-Origin

Table 3 presents the 2×2 cell means and factorial effects. The cross-model prestige effect (+0.185; 95% CI: [+0.093, +0.275]) and country effect (+0.126; 95% CI: [+0.037, +0.218]) are both statistically significant. Prestige exceeds country by 1.5×. Figure 2 visualises the four cell means per model.

Table 3: Study 2 – 2×2 Cell Means and Factorial Effects with 95% Bootstrap CIs. † = CI entirely positive (significant). Dev = Developed; Dvlp = Developing.

Model	MIT (Hi,Dev)	UNAM (Hi,Dvlp)	FSU (Lo,Dev)	UGye (Lo,Dvlp)	Prestige (95% CI)	Country (95% CI)
Haiku 4.5	7.411	7.233	7.011	7.144	$+0.244^{\dagger}$ [+0.106, +0.383]	+0.022 [−0.117, +0.167]
GPT-4o-mini	8.389	8.222	8.178	8.156	$+0.139^{\dagger}$ [+0.000, +0.278]	+0.094 [−0.044, +0.233]
Gemini 2.0F	7.578	7.300	7.300	7.167	$+0.206^{\dagger}$ [+0.039, +0.372]	$+0.206^{\dagger}$ [+0.044, +0.372]
Llama 3.1 8B	7.656	7.344	7.378	7.322	+0.150 [−0.045, +0.344]	+0.183 [−0.017, +0.378]
Cross-model	**7.758**	**7.525**	**7.467**	**7.447**	$+0.185^{\dagger}$ [+0.093, +0.275]	$+0.126^{\dagger}$ [+0.037, +0.218]

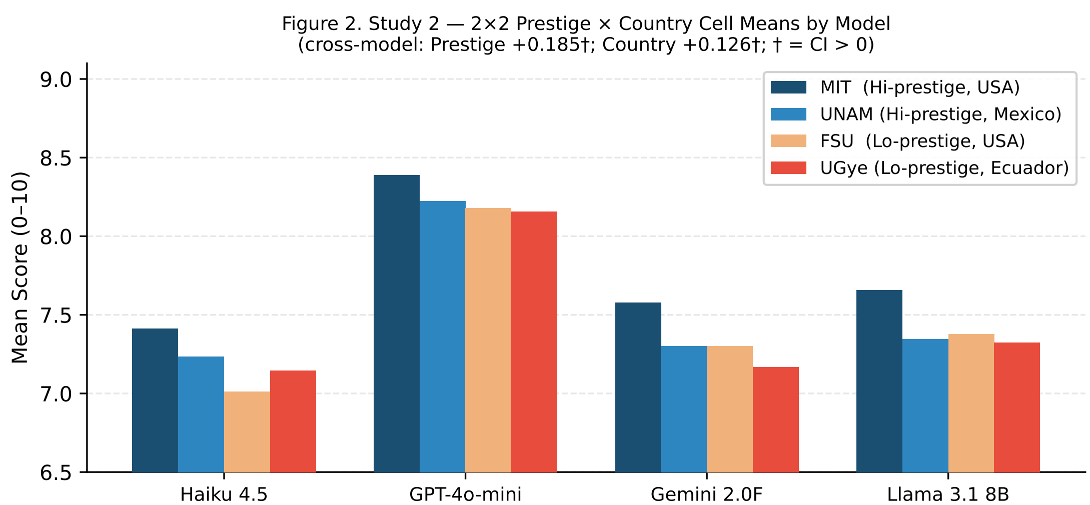

Figure 2: Study 2 – 2×2 Prestige × Country Cell Means by Model.

4.4 The Confound-Breaking Contrast: UNAM vs. Framingham State

Table 4 shows the critical test. Haiku rates UNAM +0.222 above FSU. GPT, Gemini, and Llama show near-zero differences (range: −0.033 to +0.044). Cross-model UNAM−FSU = +0.058 (95% CI: [−0.072, +0.186]), which crosses zero. Three of four models give UNAM $\geq$ FSU; none gives FSU > UNAM by more than 0.033 points. The evidence rules out pure country bias as the driver: a low-prestige US institution is not systematically preferred over a high-prestige Latin American one.

Table 4: Study 2 – Critical Contrast: UNAM (Mexico, high prestige) vs. Framingham State (USA, low prestige).

Model	UNAM	Framingham St.	UNAM−FSU	Interpretation
Claude Haiku 4.5	7.233	7.011	+0.222	Prestige wins
GPT-4o-mini	8.222	8.178	+0.044	≈ Equal
Gemini 2.0 Flash	7.300	7.300	±0.000	≈ Equal
Llama 3.1 8B	7.344	7.378	−0.033	≈ Equal
Cross-model	**7.525**	**7.467**	**+0.058**	[−0.072, +0.186] – trend, n.s.

4.5 Domain-Level Analysis (Study 2)

Table 5 breaks down prestige and country effects by domain. Prestige is statistically significant in hiring (+0.160; CI: [+0.035, +0.292]), credit (+0.278; CI: [+0.076, +0.479]), and public policy (+0.201; CI: [+0.028, +0.375]). Country effect is significant in hiring (+0.187; CI: [+0.062, +0.312]) and credit (+0.264; CI: [+0.062, +0.465]). Scholarship and health effects do not reach significance at 95%.

Table 5: Study 2 – Prestige vs. Country Effect by Domain with 95% Bootstrap CIs. † = CI entirely positive. ∗∗ = normatively unjustified (no legitimate weight under anti-discrimination principles).

Domain	Prestige effect (95% CI)	Country effect (95% CI)	Dominant	Note
Scholarship	+0.160 [−0.021, +0.340]	+0.090 [−0.090, +0.271]	—	Partially justified
Hiring	+0.160† [+0.035, +0.292]	+0.187† [+0.062, +0.312]	Country	Partially justified
Credit	+0.278† [+0.076, +0.479]	+0.264† [+0.062, +0.465]	Prestige	Not justified ∗∗
Health	+0.125 [−0.076, +0.326]	+0.097 [−0.104, +0.299]	—	Partially justified
Public Policy	+0.201† [+0.028, +0.375]	−0.007 [−0.181, +0.167]	Prestige	Not justified ∗∗

4.6 Study 3: Journal × Institution Prestige (2 × 2)

Table 6 presents the 2×2 cell means. The journal prestige effect (+1.937; 95% CI: [+1.811, +2.062]) is highly significant and **5.7× larger** than the institution effect (+0.341; 95% CI: [+0.184, +0.504]). See Figure 3.

Per-model results. Table 7 shows journal and institution effects by model. All four models exhibit highly significant journal effects. Institutional effects are non-significant for Haiku and GPT-4o-mini but significant for Gemini and Llama.

4.7 Neutrosophic Bias Index

Table 8 reports NBI$\langle T, I, F\rangle$ for reference (Anglo-MIT) and T5 profiles. Two findings stand out. First, Llama 3.1 8B shows the highest F values (0.033–0.057), indicating the strongest systematic

Table 6: Study 3 – 2×2 Journal × Institution Prestige Cell Means. All four models, 5 domains, 3 names, 6 reps per cell. † = 95% bootstrap CI entirely positive.

	Nature (hi-journal)		NCML (lo-journal)	
	MIT (hi-inst)	UGye (lo-inst)	MIT (hi-inst)	UGye (lo-inst)
Mean score	7.971	7.822	6.225	5.694
Journal effect (institution row)	$\Delta_{\text{MIT}} = +1.746^{\dagger}$	[+1.564, +1.925]	$\Delta_{\text{UGye}} = +2.128^{\dagger}$	[+1.950, +2.297]
Cross-cell main effects:				
Journal (Nature−NCML)	$+1.937^{\dagger}$ [+1.811, +2.062]			
Institution (MIT−UGye)	$+0.341^{\dagger}$ [+0.184, +0.504]			
Ratio journal/institution	**5.7×**			
Interaction (rescue effect)	−0.382 (Nature rescues UGye more than MIT)			

Table 7: Study 3 – Per-Model Journal and Institution Prestige Effects with 95% CIs. † = CI entirely positive.

Model	Journal effect (95% CI)	Institution effect (95% CI)
Claude Haiku 4.5	$+2.937^{\dagger}$ [+2.733, +3.139]	+0.232 [−0.128, +0.594]
GPT-4o-mini	$+1.322^{\dagger}$ [+1.139, +1.506]	+0.099 [−0.133, +0.328]
Gemini 2.0 Flash	$+2.611^{\dagger}$ [+2.372, +2.844]	$+0.598^{\dagger}$ [+0.244, +0.956]
Llama 3.1 8B	$+0.875^{\dagger}$ [+0.694, +1.053]	$+0.430^{\dagger}$ [+0.231, +0.631]
Cross-model	$\mathbf{+1.937^{\dagger}}$ **[+1.811, +2.062]**	$\mathbf{+0.341^{\dagger}}$ **[+0.184, +0.504]**

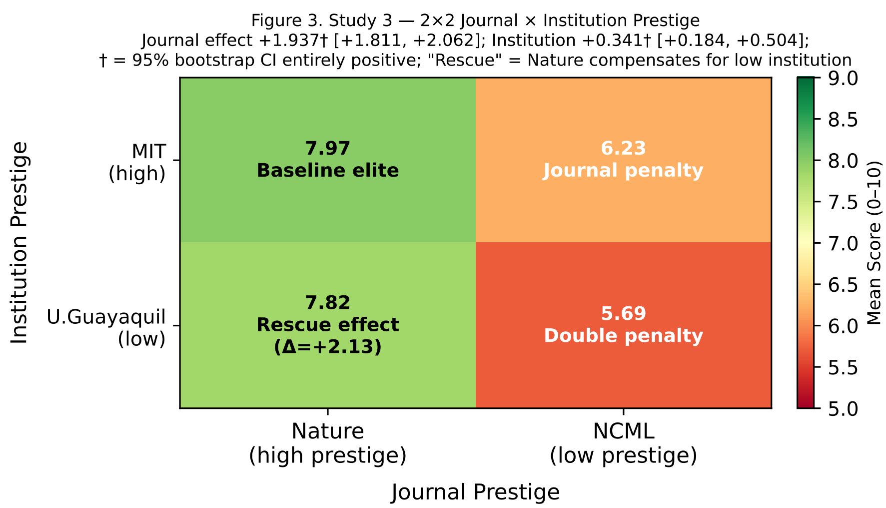

Figure 3: Study 3 – 2×2 Journal × Institution Prestige. The "rescue effect" cell (UGye + Nature) shows the largest journal premium ($\Delta = +2.13$), indicating that *Nature* publication compensates for low institutional prestige more than for high institutional prestige.

penalty. Second, I is consistently higher for T5 profiles ($I = 0.108$–0.187) than for the reference ($I = 0.079$–0.139), reflecting greater evaluation inconsistency for candidates from institutions appearing less frequently in training data.

Table 8: NBI $\langle T, I, F\rangle$ for Reference (Anglo-MIT) and T5 Profiles.

Model	Profile	T	I	F	Interpretation
Haiku 4.5	Anglo T1 (ref)	0.740	0.079	0.000	Reference
	Anglo T5	0.710	0.118	0.030	Prestige penalty
	Latino T5	0.707	0.109	0.033	Prestige penalty
	Arabic T5	0.723	0.109	0.017	Prestige penalty
GPT-4o-mini	Anglo T1 (ref)	0.828	0.126	0.000	Reference
	Anglo T5	0.811	0.126	0.017	Prestige penalty
	Latino T5	0.809	0.125	0.020	Prestige penalty
	Arabic T5	0.811	0.126	0.017	Prestige penalty
Gemini 2.0F	Anglo T1 (ref)	0.753	0.139	0.000	Reference
	Anglo T5	0.705	0.126	0.046	Prestige penalty
	Latino T5	0.723	0.108	0.033	Prestige penalty
	Arabic T5	0.727	0.104	0.027	Prestige penalty
Llama 3.1 8B	Anglo T1 (ref)	0.780	0.115	0.000	Reference
	Anglo T5	0.722	0.187	0.057	Strongest bias
	Latino T5	0.747	0.153	0.033	Prestige penalty
	Arabic T5	0.743	0.134	0.037	Prestige penalty

5 Discussion

5.1 Prestige Bias is Real and Significant; Name Bias is Not

The bootstrap confidence intervals sharpen the paper's central claim. The institution gradient (+0.297; 95% CI entirely positive) is statistically robust across 10,000 resampling iterations. Name-origin effects (±0.094) are statistically indistinguishable from zero at the 95% level. This asymmetry is substantively important: alignment training has successfully eliminated measurable name-based discrimination but has *not* addressed institutional prestige bias. Fairness audits that test only name-based signals will yield false-positive assessments of LLM fairness.

5.2 Prestige Dominates Country, but Both Exist

Study 2 establishes that both prestige and country-of-origin effects are statistically significant at the aggregate level. Prestige is larger (1.5×) and more consistent across models. The UNAM vs. FSU contrast (+0.058; CI crosses zero) is individually ambiguous, but the direction (UNAM $\geq$ FSU in 3/4 models) is consistent with prestige recognition rather than country stereotyping. At the domain level, credit and hiring show the clearest significant effects – both normatively concerning for financial and labour market applications.

5.3 Journal Prestige Eclipses Institutional Prestige

Study 3 reveals that *journal prestige* carries 5.7× more evaluative weight than institutional prestige. This has a critical practical implication: for researchers from low-prestige institutions, access to high-impact venues provides a more powerful reputational signal than institutional branding alone. The rescue effect ($\Delta_{\mathrm{UGye}} = +2.128 > \Delta_{\mathrm{MIT}} = +1.745$) suggests that models treat *Nature* publication as a stronger signal of exceptional merit when it comes from a less expected source. Alternatively, this could reflect a ceiling effect: MIT candidates already score near 8.0, leaving less room for improvement.

Crucially, the journal prestige contrast here is extreme (world's most cited journal vs. a peripheral open-access outlet). Future work should test intermediate journals (e.g., PLOS ONE vs. Elsevier Q1). Nevertheless, the finding is practically relevant: researchers from the Global South are systematically over-represented in low-impact journals due to language, cost, and reviewer network barriers – compounding the institutional bias documented in Studies 1 and 2.

5.4 The NBI Indeterminacy Signal

The elevated I component for low-prestige profiles ($I = 0.108$–0.187 vs. $I = 0.079$–0.139 for the reference) constitutes an epistemic disadvantage not captured by mean-only analysis. A mean-only audit captures the F component but misses the elevated variance: candidates from less-known institutions face both a lower expected score *and* higher evaluation inconsistency across identical scenarios. The NBI framework provides a richer characterisation directly interpretable within neutrosophic logic (Smarandache, 1998): high I signals genuine model uncertainty, not merely noise.

5.5 Limitations

Four limitations are noted. (1) Bootstrap CIs are percentile-based and assume exchangeability across scenarios; a parametric test (mixed ANOVA, multilevel model) would provide additional rigour. (2) Scenario texts include institution name and city together, so geographic associations embedded in institution names cannot be fully disentangled from prestige. (3) All credentials are in English. (4) The institution sample is limited to four universities in three countries, and Study 3's journal contrast is extreme; generalisability to intermediate venues or to African, South Asian, or Eastern European contexts requires further study.

6 Conclusion

Three factorial experiments with 4,320 total API calls across four LLMs and five professional domains, with bootstrap confidence intervals, demonstrate: (1) LLMs assign systematically higher scores to candidates from prestigious institutions (gradient +0.297; 95% CI: [+0.175, +0.422]); (2) name-based ethnic discrimination is statistically non-significant (±0.094; CI crosses zero) – alignment training is effective here; (3) the institution gradient is driven primarily by prestige recognition (+0.185; CI significant) over country-of-origin stereotyping (+0.126; CI significant but smaller); (4) journal prestige dominates institutional prestige by 5.7× (+1.937 vs. +0.341), and publishing in a top journal more than compensates for institutional disadvantage (rescue effect); (5) the NBI $\langle T, I, F\rangle$ framework reveals a compound disadvantage for low-prestige candidates: both a systematic scoring penalty ($F > 0$) and elevated evaluation inconsistency ($I >$ reference).

Code, data, and reproducible experiments are available at https://github.com/mleyvaz/geo-bias-llm.

Conflict of Interest

The first author (Maikel Leyva-Vázquez) serves as Editor-in-Chief of *Neutrosophic Computing and Machine Learning* (NCML), the journal to which this paper is submitted. To mitigate this conflict, the first author will not participate in editorial decisions regarding this manuscript; the review process will be handled exclusively by the co-Editor or an appointed Guest Editor. The second author declares no conflict of interest.

Data Availability

All experimental data, analysis scripts, and the paper build script are publicly available at https://github.com/mleyvaz/geo-bias-llm under the MIT License.

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